

# Automation and Monitoring System

## Presentation Overview of System Capabilities

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# What the Monitoring System Offers

- Continuous observability for the deployed machine learning endpoint.
- Automatic capture of live inference traffic for later inspection and analysis.
- Performance monitoring through latency, invocation, and error metrics.
- Feature drift detection by comparing live data against baseline training behavior.
- Centralized dashboards for quick operational review.
- Email alerts when endpoint behavior becomes abnormal.

# End-to-End Monitoring Architecture

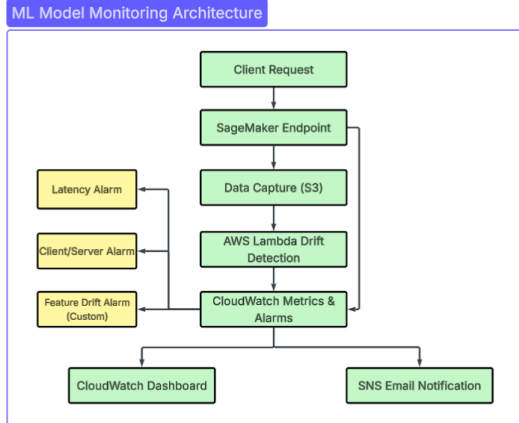


Figure: Monitoring architecture for endpoint observability, drift detection, and alerting.

# Core Services and Their Roles

| Component          | What it provides                                              |
|--------------------|---------------------------------------------------------------|
| SageMaker Endpoint | Hosts the model and serves real-time inference requests.      |
| S3 Data Capture    | Stores request and response payloads from endpoint traffic.   |
| AWS Lambda         | Analyzes captured data and produces custom drift metrics.     |
| CloudWatch         | Tracks metrics, visualizes system health, and manages alarms. |
| Amazon SNS         | Delivers email notifications when alarms are triggered.       |

# Endpoint Health Monitoring

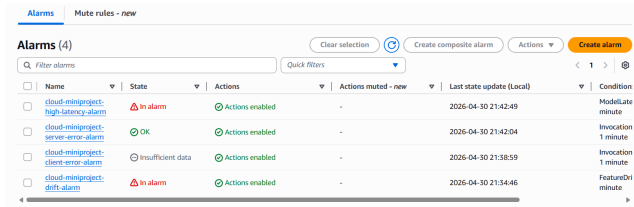
- **Model latency:** identifies slow inference responses and possible performance degradation.
- **Invocation count:** shows request volume and endpoint usage patterns.
- **4XX errors:** highlights invalid or malformed client requests.
- **5XX errors:** detects server-side failures or model container problems.
- **Alarm states:** summarize whether the endpoint is operating normally or requires attention.

# Feature Drift Detection

- Compares live inference features with baseline training distributions.
- Monitors financial indicators such as moving averages, RSI, MACD, MACD signal, and MACD histogram.
- Produces a custom drift score that summarizes how far live traffic has shifted from expected behavior.
- Publishes the drift score to CloudWatch so it can be visualized, tracked, and alarmed on.
- Helps identify when model predictions may become less reliable due to changing market conditions.

# Automated Alerts

- CloudWatch alarms continuously evaluate endpoint and drift metrics.
- Abnormal latency, invocation failures, or high drift can trigger an alarm.
- Amazon SNS sends email notifications to subscribed users.
- Operators receive timely visibility into issues without manually checking dashboards.



The screenshot shows the AWS CloudWatch Alarms console. At the top, there are tabs for 'Alarms' and 'Mute rules - new'. Below the tabs, there's a section for 'Alarms (4)' with a search bar and a 'Quick filters' dropdown. To the right of the search bar are buttons for 'Clear selection', 'Create composite alarm', 'Actions', and a prominent orange 'Create alarm' button. Below this is a table listing four alarms. Each row has a checkbox, the alarm name (with a link), its state (with an icon), its actions (with a link), whether it's muted, the last state update (local), and its condition.

| <input type="checkbox"/> | Name                                                 | State                            | Actions                         | Actions muted - new | Last state update (Local) | Condition           |
|--------------------------|------------------------------------------------------|----------------------------------|---------------------------------|---------------------|---------------------------|---------------------|
| <input type="checkbox"/> | <a href="#">cloud-miniproject-high-latency-alarm</a> | <span>⚠ In alarm</span>          | <a href="#">Actions enabled</a> | -                   | 2026-04-30 21:42:49       | ModelLatency minute |
| <input type="checkbox"/> | <a href="#">cloud-miniproject-server-error-alarm</a> | <span>✅ OK</span>                | <a href="#">Actions enabled</a> | -                   | 2026-04-30 21:42:04       | Invocation 1 minute |
| <input type="checkbox"/> | <a href="#">cloud-miniproject-client-error-alarm</a> | <span>⚠ Insufficient data</span> | <a href="#">Actions enabled</a> | -                   | 2026-04-30 21:38:59       | Invocation 1 minute |
| <input type="checkbox"/> | <a href="#">cloud-miniproject-drift-alarm</a>        | <span>⚠ In alarm</span>          | <a href="#">Actions enabled</a> | -                   | 2026-04-30 21:34:46       | FeatureDrift minute |

# Monitoring Dashboard

cloud\_mini\_project\_dashboard ▾ ☆

1h

3h

12h

1d

3d

1w

Custom

Local timezone ▾



10 seconds ▾

Esc

Actions ▾

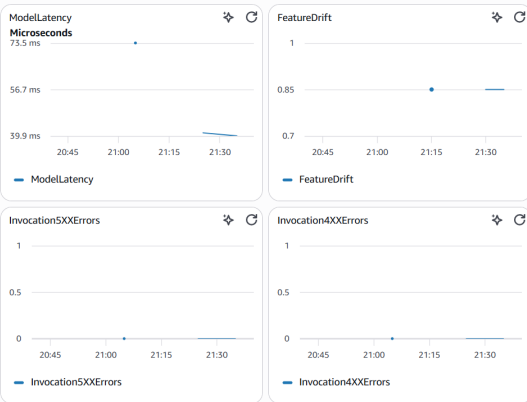


Figure: Dashboard view for endpoint performance, errors, usage, and drift metrics.



- **Reliability:** detects endpoint failures and performance degradation early.
- **Transparency:** makes endpoint behavior visible through metrics and dashboards.
- **Model awareness:** tracks whether live data still resembles training data.
- **Faster response:** notifies users automatically when intervention may be needed.
- **Extensibility:** supports future additions such as scheduled drift checks, confidence monitoring, and retraining workflows.

## The system offers

A cloud-based monitoring layer that captures inference traffic, tracks endpoint health, detects feature drift, visualizes operational metrics, and sends automated alerts when abnormal behavior appears.